

Disk Arthroplasty, Cervical

ACG: A-0227 (AC)

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MCG Health
Ambulatory Care
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Clinical Indications for Procedure

- Disk arthroplasty may be indicated when **ALL** of the following are present(1)(2)(3)(4)(5):
 - Age 18 years or older(8)(9)
 - Cervical degenerative disk disease between C3 and C7, and placement of 1 or 2 contiguous-level artificial disks is planned(9)(10)(11)(12)(13)
 - Symptoms corresponding to level(s) at which disk degeneration has been identified, as indicated by **1 or more** of the following(10)(11):
 - Cervical radiculopathy (ie, arm or neck pain, paresthesia)
 - Cervical spondylotic myelopathy (ie, upper extremity weakness, paresthesia, loss of balance, neck pain)(20)
 - Failure of nonoperative treatment (eg, NSAIDs, narcotic or non-narcotic analgesics, physical therapy, spinal manipulation therapy)(21)(22)
 - No evidence of cervical instability, as indicated by **1 or more** of the following:
 - Absence of sagittal plane angulation of more than 11 degrees on lateral flexion-extension x-rays(23)
 - Absence of sagittal plane translation of more than 3 mm on lateral flexion-extension x-rays(10)(21)
 - No anatomic deformity or malignancy at affected level(s)(8)
 - No inflammatory spondyloarthropathy (eg, ankylosing spondylitis, rheumatoid arthritis)(24)
 - No osteoporosis(8)(10)(11)(24)
 - No previous cervical surgery at affected level(11)(24)
 - No systemic infection or localized infection at site of implantation(8)(11)(24)

Alternatives to Procedure

- Alternatives include(22)(25)(26):
 - Anterior cervical fusion
 - Cervical discectomy or microdiscectomy
 - Cervical orthosis(27)
 - Patient education in self-management and exercise
 - Physical therapy. See Spine Soft Tissue Dysfunction Rehabilitation [AC](#) for further information.
 - Spinal manipulation therapy. See Spinal Manipulation Therapy (SMT), Chiropractic and Other [AC](#) for further information.

Evidence Summary

Background

Disk arthroplasty (artificial intervertebral disk implantation or replacement) was developed as an alternative to spinal fusion to maintain spinal segment mobility and theoretically prevent subsequent disk degeneration of adjacent segments, which is a complication of spinal fusion.(1)(4)(6)(7) (**EG 2**)

Criteria

The evidence for the clinical indications found in this guideline includes 22 published peer reviewed articles.

For cervical degenerative disk disease between C3 and C7, when placement of 1 or 2 contiguous-level artificial disks is planned, evidence demonstrates at least moderate certainty of at least moderate net benefit. (**RG A1**) A systematic review of 8 studies (6 randomized trials) evaluating the efficacy of anterior cervical disk arthroplasty compared with anterior cervical decompression fusion in patients with radiculopathy found, at 2-year follow-up, no difference in neck pain (as measured by visual analog scale or numeric rating scale) between groups; of the 6 studies that measured Neck Disability Index (NDI) scores, one study found that anterior cervical decompression fusion was associated with a statistically significant, but not clinically meaningful, improvement compared with anterior cervical disk arthroplasty. The authors noted that heterogeneity between studies and lack of assessor blinding limited the results; further high-quality studies with larger study populations, double blinding, and specific inclusion criteria to allow for generalizability were recommended.(13) (**EG 1**) A systematic review and meta-analysis of 22 studies including 14 randomized controlled trials (3160 patients) that compared anterior cervical disk arthroplasty with anterior cervical discectomy and fusion (the majority of procedures were for single-level disease) found, at a follow-up range of 4 to 7 years, that cervical disk arthroplasty was associated with neurologic success and overall success; the authors concluded that anterior cervical disk arthroplasty is not inferior to anterior cervical discectomy and fusion and may, in fact, be superior to the traditional procedure for some outcomes.(14) (**EG 1**) A health technology assessment based upon a systematic review of 8 studies found that single-level cervical disk arthroplasty is an effective and safe alternative to anterior cervical discectomy and fusion, providing statistically significant reductions in pain and disability.(15) (**EG 1**) Follow-up data from one of the studies included in the health technology assessment (128 patients with arthroplasty and 104 patients with fusion) found, at 10 years post procedure, that arthroplasty was associated with improved NDI scores compared with fusion, with no difference in neurologic success (defined as maintenance or improvement of motor or sensory function) or symptomatic adjacent-level disease requiring surgery between the groups.(16) (**EG 1**) A systematic review and meta-analysis of 9 randomized controlled and 2 clinical controlled trials (2715 patients) comparing anterior cervical disk replacement with anterior cervical decompression and fusion for the treatment of 2-level cervical spondylosis found, at a follow-up range of 18 to 84 months, that treatment with anterior cervical disk replacement was associated with favorable outcomes as assessed by NDI, visual analog scale for neck pain, and Short Form-12 physical component scores; anterior cervical disk replacement was also positively associated with overall clinical success, device-related complications, and subsequent surgical intervention. The authors recommended additional large randomized studies to evaluate this intervention.(17) (**EG 1**) A meta-analysis of 41 studies (including 27 randomized trials) with a total of 7532 patients comparing cervical disk arthroplasty with anterior cervical discectomy and fusion found that cervical disk arthroplasty was associated with fewer adjacent segment reoperations, a lower rate of adjacent segment degeneration (defined as radiologic degeneration) or adjacent segment disease (defined as radiologic degeneration with symptoms), and greater cervical range of motion compared with anterior cervical discectomy and fusion.(18) (**EG 1**) A multicenter randomized trial of 153 patients with cervical radiculopathy compared disk arthroplasty with fusion surgery and found, at a mean follow-up of 66 months, no difference in NDI or quality-of-life scores between the groups.(19) (**EG 1**)

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment across patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 422.101(28); 42 CFR 419.22(29)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(30); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(31); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(32)

Medicare Coverage Determinations: Medicare Coverage Database(33)

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